

MONROE CUSTER, MI, USA

WMO: 725418

Lat: 41.940N

Lon: 83.435W

Elev: 187

StdP: 99.10

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54823

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-13.9	-22.0	0.5	-17.1	-17.9	0.8	-12.9	11.6	-2.5	10.7	-3.0	2.6	240	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.4	23.4	31.1	23.0	29.0	22.1	25.5	29.6	24.5	28.5	23.6	27.4	4.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.4	27.4	23.0	18.2	26.1	22.5	17.6	25.7	79.1	29.5	74.7	28.5	71.5	27.9	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.7	7.7	-20.8	34.5	3.5	1.4	-23.3	35.5	-25.4	36.4	-27.3	37.1	-29.9	38.2		
(5)			-21.2	27.1	3.4	1.2	-23.6	27.9	-25.6	28.6	-27.4	29.2	-29.9	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-3.3	-2.3	2.7	9.4	15.7	20.9	22.8	21.6	18.2	11.7	5.5	-0.3		
	DBStd	10.43	6.37	5.93	5.70	4.91	4.69	3.67	3.11	2.88	3.99	4.81	5.12	5.58		
	HDD10.0	1580	412	345	236	69	6	0	0	0	1	36	153	322		
	HDD18.3	3362	670	577	484	272	110	16	2	6	51	212	385	578		
	CDD10.0	1684	1	1	11	49	183	328	396	359	247	90	18	2		
	CDD18.3	424	0	0	0	3	28	93	139	106	47	7	0	0		
	CDH23.3	3718	0	0	4	40	264	825	1291	856	384	53	1	0		
	CDH26.7	1131	0	0	0	4	72	280	435	224	106	9	0	0		
(14) Wind	WSAvg	3.2	3.9	3.7	3.6	3.9	3.2	2.9	2.3	2.0	2.3	3.0	3.4	3.7		
(15) Precipitation	PrecAvg	847	50	48	61	83	86	88	83	85	80	61	65	59		
	PrecMax	1192	91	130	140	128	147	203	163	200	158	156	167	153		
	PrecMin	642	14	11	14	29	20	43	32	16	14	12	20	16		
	PrecStd	125	22	29	28	28	41	35	29	39	36	34	30	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	16.7	22.7	27.4	31.2	33.8	34.1	32.6	32.4	27.8	21.2	16.1		
		MCWB	13.1	12.5	16.8	18.1	21.5	22.7	24.5	24.2	23.3	20.1	15.1	14.1		
	2%	DB	11.3	12.4	17.7	23.9	28.7	32.2	32.5	31.1	29.5	24.4	17.8	12.5		
		MCWB	10.0	10.3	13.5	16.1	20.5	22.5	23.9	23.8	21.9	18.4	13.7	10.9		
	5%	DB	7.7	7.9	14.1	21.2	26.9	29.9	31.1	28.9	27.4	22.4	16.0	9.1		
		MCWB	6.1	5.8	10.9	14.8	19.5	21.7	23.5	22.9	20.8	17.1	12.4	7.6		
	10%	DB	4.9	5.0	11.9	17.8	23.8	27.7	28.9	27.6	25.7	20.0	13.0	7.2		
		MCWB	3.5	3.1	9.0	12.8	18.3	20.5	22.6	21.7	20.0	16.2	10.4	5.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	13.6	17.5	19.5	23.6	25.2	27.2	26.3	25.4	22.5	16.9	14.1		
		MCDB	14.6	15.5	21.5	24.9	28.0	30.6	30.4	30.2	29.8	25.8	19.4	15.8		
	2%	WB	10.3	10.2	14.4	17.5	22.0	24.0	25.8	25.2	23.8	20.2	14.8	11.3		
		MCDB	11.1	12.3	17.3	21.6	26.5	28.9	29.9	28.6	27.3	22.9	17.3	12.5		
	5%	WB	6.4	6.1	11.8	15.5	20.6	23.2	24.7	24.1	22.5	18.2	12.9	7.9		
		MCDB	7.6	7.8	14.0	20.1	24.8	27.5	28.9	27.4	25.6	20.7	15.1	8.9		
	10%	WB	3.5	3.3	8.7	13.5	19.3	22.2	23.7	23.1	21.2	16.6	11.0	5.6		
		MCDB	4.7	4.7	11.3	17.8	23.0	26.3	27.4	26.2	24.0	19.3	13.2	7.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	7.8	8.8	10.4	10.5	10.6	11.0	10.7	11.3	10.4	8.5	6.5		
		MCDBR	9.4	11.0	13.5	14.9	13.2	12.7	12.1	11.8	13.3	12.9	11.9	9.2		
	5% WB	MCWBR	8.5	9.3	9.7	8.8	6.6	5.5	5.4	5.9	6.9	7.4	8.8	8.1		
		MCWBR	9.0	10.7	12.9	13.1	11.4	10.7	10.5	10.0	10.6	10.7	10.8	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.337	0.358	0.404	0.435	0.452	0.443	0.435	0.389	0.361	0.322	0.314		
	taud		2.407	2.287	2.324	2.225	2.186	2.186	2.231	2.241	2.346	2.406	2.478	2.401		
	Ebn at Noon		845	859	885	865	844	828	831	826	842	825	816	792		
	Edh at Noon		79	104	114	135	144	145	137	132	110	92	74	74		
(44) All-Sky Solar Radiation	RadAvg		1.53	2.42	3.50	4.62	5.40	6.05	6.08	5.30	4.30	2.82	1.78	1.24		
	RadStd		0.15	0.31	0.33	0.40	0.47	0.40	0.32	0.20	0.36	0.23	0.24	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.78	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86	N/A

Nomenclature: See separate page